

The Next-Level FPGA ARCHITECTURES



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This article discusses what to expect from next-level field programmable gate arrays

Architectures of different field programmable gate arrays (FPGA) differ from vendor to vendor. Some vendors are Silego, Lattice Semiconductor, Xilinx, Microsemi and Achronix. Silego uses mixed-signal components. Whereas, Lattice Semiconductor is known for small form factors and ultra-low power FPGAs. Lattice Semiconductor's iCE40UP3K FPGAs are designed for ultra-low power mobile applications, which include smartphones, tablets and other handheld devices.

Microsemi has SmartFusion2 system on chip (SoC) FPGAs. Altera and Xilinx FPGAs have great capacity and performance. Achronix Semiconductor's asynchronous FPGAs operate at multi-gigahertz frequencies while consuming relatively little power. Achronix's Speedcore embedded FPGAs improve throughput and latency performance by 10 times, and reduce power by 50 per cent and cost by 90 per cent.

Silego FPGA

Silego announced its next-generation GPAK5 family, featuring a number of new and enhanced functionalities, including zero static-power asynchronous state machine (ASM) block, 8x8 scratchpad memory and I2C block. These new func-

tions, along with configurable digital and analogue fabric seen in previous GPAK generations, are presented in a tiny 2.0mm x 3.0mm 20-pin STQFN package with 18 general-purpose inputs/outputs (GPIOs).

Lattice Semiconductor FPGA

The iCE40 Ultra Plus family includes integrated SPI and I2C blocks to interface virtually with all mobile sensors and application processors. iCE40UP3K features two-device density of 2800 range look-up tables (LUTs) with programmable IOs, which can be used as either SPI or I2C interface ports or GPIOs.

It also features 20 embedded block RAM memory blocks, 80 embedded block RAM memory bits, on-chip oscillators and four DSP blocks.

iCE40UP is available in a tiny form factor package—as small as 2.15mm x 2.55mm. It allows the device to easily fit into many applications. Typical applications include iCE40 Ultra devices, service LEDs, GPIO expanders, serial data IO level shifters and voice recognition for mobile applications.

Features include:

1. 2800 two-device logic architecture
2. 40nm low power processors
3. 100µA standby current
4. 1024kb single-port SRAM
5. Two IO pins to support I3C interface
6. Two hardened SPI interfaces
7. 24mA current drive RGB LED outputs
8. 2.15mm x 2.55mm ultra-small form factor
9. On-chip DSP
10. WLCS and QFN packages

Applications include voice-recognition applications, smartphones, tablets and consumer handheld devices, commercial and industrial handheld devices, multi-sensor management applications, sensor pre-processing and sensor fusion, sensor applications and USB 3.1 type-C cable detection/power delivery, among others.

Microsemi FPGA

SmartFusion2 SoC FPGAs offer 5000 to 150,000 logic elements (LEs) with a 166MHz ARM Cortex-M3 processor, including ETM

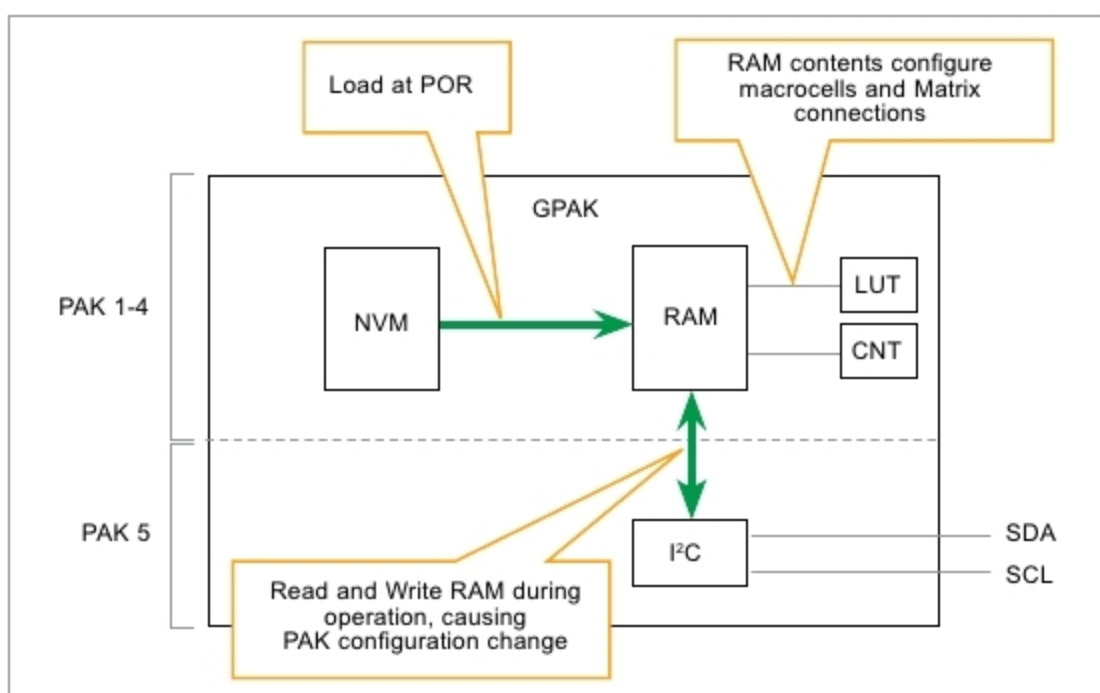


Fig. 1: GPAK5 tiny mixed-signal FPGA featuring ASM and I2C